

Program: Diploma in Integrated Circuit Design and Fabrication	
Course Code: 5372A	Course Title: Analog and Digital Communication Systems
Semester: 5	Credits: 3
Course Category: Program Elective Course	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To introduce the concept of analog and digital communication systems.
- To enrich on digital pulse modulation techniques, digital modulation schemes.

Course Prerequisites:

Topic	Course code	Course name	Semester
Multiplexer, binary codes	3044	Digital Electronics	3
Knowledge of electronics components	2031	Fundamentals of Electrical and Electronics engineering	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain various parameters of Amplitude Modulation (AM), Frequency Modulation (FM) and analog pulse modulation techniques.	11	Understanding
CO2	Explain the working of AM and FM transmitters & receivers.	11	Understanding
CO3	Explain various digital pulse modulation techniques	11	Understanding
CO4	Summarize digital modulation schemes, multiplexing techniques and data transmission methods.	10	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain various parameters of Amplitude Modulation (AM), Frequency Modulation (FM) and analog pulse modulation techniques.		
M1.01	Classify modulation techniques	1	Understanding
M1.02	Outline modulation parameters of different AM techniques	4	Understanding
M1.03	Derive the expression for modulated signals and frequency spectrum of FM	3	Understanding
M1.04	Explain Analog Pulse Modulation Schemes	3	Understanding
Concept of modulation. Amplitude Modulation: Definition, Expression of AM wave, Modulation index, AM wave representation, Bandwidth and power calculation, Frequency spectrum of AM, DSBSC, SSB and VSB. Frequency Modulation: Definition, Mathematical representation of FM, Modulation index, Frequency spectrum of FM. Comparison of AM and FM, Applications of AM and FM. Analog Pulse Modulation Schemes: PAM, PWM, PPM-Block diagram -Explanation-Waveform.			
CO2	Explain the working of AM and FM transmitters & receivers		
M2.01	Explain the working of AM transmitter	3	Understanding
M2.02	Explain the working of FM transmitter	3	Understanding
M2.03	Explain the characteristics of radio receiver	1	Understanding

M2.04	Illustrate superheterodyne receiver, AM and FM receivers	4	Understanding
	Series Test - I	1	
Contents: AM transmitter: AM Transmitter - low level and high level, AM Balanced modulator (Block level Explanation Only). FM transmitter: FM Transmitter-Armstrong method (Block level Explanation Only), Pre-Emphasis and De-Emphasis circuits Characteristics of radio receiver - selectivity, sensitivity, fidelity and noise figure. (Definition Only). Superheterodyne receiver, AM receiver (block level explanation only), FM receiver (block level explanation only)			
CO3	Explain various digital pulse modulation techniques		
M3.01	Outline digital communication systems	1	Understanding
M3.02	Explain various types of sampling and quantization	3	Understanding
M3.03	Interpret the PCM systems	3	Understanding
M3.04	Explain DPCM, DM and ADM	4	Understanding
Contents: Necessity of digital communication systems, Block diagram and sub-system description of a digital communication system Sampling and Quantization: Definition of sampling, Types of sampling techniques, Sampling theorem and Nyquist rate, Definition of quantization, Uniform and Non-uniform Quantization, Quantization error. Basic elements of PCM, Block diagram of PCM, Quantization noise in PCM, Bandwidth Requirement, Commanding. (Simple problems) DPCM, Delta Modulation, Noise in Delta Modulation: slope overload and granular noise, Adaptive Delta modulation			
CO4	Summarize digital modulation schemes, multiplexing techniques and data transmission methods.		
M4.01	Summarize digital modulation schemes.	4	Understanding
M4.02	Compare various multiplexing techniques	3	Understanding

M4.03	Outline data transmission methods	3	Understanding
	Series Test – II	1	
Contents: Digital modulation schemes Generation and Reception of ASK, BFSK, BPSK, Comparison of ASK, BFSK and BPSK. Concept of DPSK, QPSK Multiplexing Techniques: TDM & FDM-Concept, Block diagram, Advantages, Disadvantages, Applications, and Comparison. Data transmission methods: Simplex, Half-duplex, Full duplex, Synchronous and Asynchronous			

Text/Reference

T/R	Book Title/Author
T1	Principles of Communication systems – Taub and Schilling
T2	Electronic Communication Systems – George Kennedy – TMH
T3	Communication Systems - Simon Haykin
R1	Electronic Communication Systems - Frank R Dungan - 3rd Edition – Thomson
R2	Digital Communication -John G. Proakis, Masoud Salehi, , McGraw Hill Education Edition, 2014
R3	Communication Systems (Analog and Digital) – Sanjay Sharma
R4	Elements of Information Theory- T Cover and J Thomas
R5	Principles of Electronic Communications –Pradeep Kumar Ghosh

Online resources:

Sl.No.	Website Link
1	https://nptel.ac.in/courses/117101051/
2	https://www.tutorialspoint.com/digital_communication
3	https://nptel.ac.in/courses/108101113/
4	https://nptel.ac.in/courses/117/105/117105143/
5	https://www.slideshare.net/bhavyaw/noise-13468777